

Lec 14: TCP introduction & TCP header

Computer Networks (CSE 232)

Rinku Shah

These slides are inspired and adapted from:

1. *Computer Networking: A Top-Down Approach*
6th edition, Jim Kurose, Keith Ross, Pearson, 2017



INDRAPRASTHA INSTITUTE of
INFORMATION TECHNOLOGY
DELHI

TCP: overview

RFCs: 793,1122, 2018, 5681, 7323

- **point-to-point:**
 - one sender, one receiver
- **reliable, in-order *byte stream*:**
 - no “message boundaries”
- **full duplex data:**
 - bi-directional data flow in same connection
 - MSS: maximum segment size
- **cumulative ACKs**
- **pipelining:**
 - TCP congestion and flow control set window size
- **connection-oriented:**
 - handshaking (exchange of control messages) initializes sender, receiver state before data exchange
- **flow controlled:**
 - sender will not overwhelm receiver

TCP segment structure

